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Alishan Forest Railway

A 762 mm mountain railway from Chiayi to Alishan, with spiral, Ω -shaped and Z-shaped climbing works, Shay locomotives and active forest-park branches.

ALISHAN FOREST RAILWAY			
OPERATOR Alishan Forest Railway and Cultural Heritage Office ^[1]	GAUGE 762 mm ^[2]	MAIN-LINE STATIONS 16 in the current published timetable ^[3]	MAIN-LINE PUBLISHED CHAINAGE 71.6 km to Alishan ^[1]
HIGHEST NAMED STATION Zhushan, 2,451 m ^[1]	PASSENGER REOPENING 6 July 2024 ^[10]		

The Alishan Forest Railway is built around a constraint that ordinary railway descriptions often hide: it climbs a high mountain on 762 mm track without a rack rail.^{[2] [9]} The operator dates the railway's opening to 1912 and describes a main line of approximately 72 km, while the current station overview measures the Chiayi–Alishan run to 71.6 km.^{[2] [1]} The Ministry of Culture's potential-site page uses 71.34 km for a wider cultural-landscape boundary, so the line does not have one source-independent “length” to substitute for the others.^[9]

The current main-line timetable is effective from 10 January 2025 and shows two daily trains up the mountain and two down, with the longer up train leaving Chiayi [\[/rail/thsr/stations/chiayi/\]](/rail/thsr/stations/chiayi/) at 10:00 and reaching Alishan at 14:56 after a scheduled Fenqihu stop.^[3] The 16 main-line names are Chiayi, Beimen [\[/rail/metro/stations/g13/\]](/rail/metro/stations/g13/), Luma Chan, Zhuqi, Muliliao, Zhangnaoliao, Duli Shan, Liyuanliao, Jiaoliping, Shuisheliao, Fenqihu, Duolin, Shizilu, Pingzhena, Erwanping and Alishan.^{[3] [8]} The timetable is a service record, not a claim that every historical station, siding or timber branch still has passenger service.

The mountain route is a sequence of controlled reversals and long curves. The official heritage description places a three-turn spiral at Duli Shan, a large horseshoe or 180-degree curve between Zhuqi and Zhangnaoliao, and Z-shaped reversals farther up the mountain where there is not enough shelf to turn a train normally.^[9] The operator's own overview adds that the railway passes eight tunnels on the Duli Shan climb and that the Ω -shaped track at Shuisheliao turns the train through 180 degrees.^[1] The published maximum gradient is 6.25%, and the source describes trains as adhesion-worked rather than rack-assisted.^[9]

Alishan is also a junction rather than a simple terminal. The operator identifies the forest-park branches as Shenmu, Zhaoping and Zhushan, with frequent short workings on the first two and a sunrise-dependent departure notice for Zhushan.^[4] Duigaoyue is a named platform and passing location on the Zhushan alignment; its presence is confirmed by the operator's station

overview, but the current branch timetable does not establish a scheduled boarding stop there. [1] [4] The Mianyue Line is a separate historical case: it was built as a timber route, later reopened as a tourist railway, and closed after the 1999 earthquake damaged its trackbed and tunnels and later repairs continued to fail. [4]

The rolling-stock story follows the railway's changing work. The operator records 18-ton two-cylinder Shay locomotives bought in eight units between 1910 and 1913 and twelve 28-ton three-cylinder locomotives bought between 1912 and 1917; it says these LIMA-built machines were selected for the line's steep, curved mountain route. [5] The official English locomotive page says Nos. 25 and 31 were repaired and returned to running, with coal bunkers converted to generator rooms, air compressors added and coal replaced by diesel fuel; it dates the repairs to 2006 and 2004 respectively. [6] That is why "steam locomotive" needs qualification here: the boiler-and-cylinder form survives, but the two operating examples described by the operator are diesel-fired after conversion. [6]

The operator's FAQ gives a different roster perspective for No. 26, identifying it as a 1912 28-ton Shay and placing it in static display, while saying that 28- and 18-ton examples are displayed at the Chiayi and Fenqihu facilities and one locomotive may run irregularly in the mountains. [8] The diesel table is the better source for the everyday fleet: it identifies active DL25–30, DL31–34, DL38, DL39–43 and DL45–51 groups, with retired early Mitsubishi and O&K groups kept in the same record so that "current" is not mistaken for "ever acquired." [7]

The line's recent service history is inseparable from typhoon and landslide work. The Forestry and Nature Conservation Agency says Typhoon Morakot and Typhoon Dujuan damaged the railway and that the collapse of No. 42 left the line interrupted for 15 years before the 6 July 2024 full-line opening. [10] The railway office's completion notice records two collapses totalling 55 m, a new 1,106 m tunnel begun on 8 January 2021, and construction challenges including water inflow, material transport and pandemic labour shortages. [11] A government recreation chronology gives 29 December 2023 as the date the last-mile construction was completed, while the passenger announcement gives 6 July 2024 as the date scheduled trains again ran from Chiayi to Alishan. [12] [10] These are different milestones and are published as such.

The 2024 reopening did not end weather-related operational risk. [13] An official October 2024 notice says the railway suspended the Chiayi–Alishan main line and the Shenmu, Zhaoping and Zhushan branches while Typhoon Krathon was in the warning period, then resumed normal service on 5 October after a route inspection found no damage. [13] That notice is evidence for a specific typhoon closure and recovery, not a general operating guarantee; later interruptions remain TBC unless the operator's dated notices are checked.

The heritage case is similarly specific. The Ministry of Culture presents the Alishan Forestry and Railway Cultural Landscape as a Taiwan World Heritage potential site and describes the 762 mm railway, its mountain methods, forest ecology and associated settlements. [9] The site is not presented here as a UNESCO-inscribed property. A separate government chronology records its domestic registration as an important cultural landscape in 2019, which is a national cultural-heritage status rather than UNESCO inscription. [12] The route, station inventory, rolling-stock roster and closure history therefore belong together, but each claim stays tied to the primary record that actually supports it.

Specifications

Main line	71.6 km ^[1]
Maximum gradient	6.25 percent ^[9]
Active branch services	Shenmu, Zhaoping and Zhushan ^[4]
Mianyue Line	Closed; 9.2 km historical tourist railway ^[4]
Replacement No. 42 tunnel	1,106 m ^[11]

References

13 of 13 sources on this page are primary — published by the operator, the builder or government.

1

Alishan Forest Railway station overview

[https://afrch.forest.gov.tw/]

阿里山百年鐵道

PRIMARY

Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處)

accessed 2026-08-25

Current operator overview for station names, chainages, elevations and branch points.

2

Railway technology

[https://afrch.forest.gov.tw/0000524]

鐵道技術

PRIMARY

Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處)

accessed 2026-08-25

Operator technical history for the 1912 opening, 762 mm gauge and mountain methods.

3

Main-line timetable

[https://afrch.forest.gov.tw/0000115]

林鐵本線

PRIMARY

Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處)

accessed 2026-08-25

Published main-line station sequence and daily service, effective 10 January 2025.

4

Branch-line timetable and closure notes

[https://afrch.forest.gov.tw/0000300]

林鐵支線

PRIMARY

Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處)

accessed 2026-08-25

Branch service records, Mianyue closure history and the 6 July 2024 branch timetable boundary.

5

Shay steam locomotive history

[https://afrch.forest.gov.tw/0000107]

夏依式蒸汽機車

PRIMARY

Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處)

accessed 2026-08-25

Operator history of the LIMA Shay classes and their role during railway construction.

6 Introduction of steam locomotive [<https://afrch.forest.gov.tw/EN/0000108>]

Introduction of Steam Locomotive

PRIMARY Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處)
accessed 2026-08-25

Official English mirror for the current No. 25 and No. 31 repair and fuel-conversion record; the Chinese page timed out during this check.

7 Introduction of diesel locomotive [<https://afrch.forest.gov.tw/0000106>]

柴油機車之介紹

PRIMARY Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處)
accessed 2026-08-25

Operator table of diesel locomotive groups, manufacturers, build years, weights and current/retired status.

8 Frequently asked questions [<https://afrch.forest.gov.tw/faqs?iPage=2>]

常見問答

PRIMARY Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處)
accessed 2026-08-25

The operator's station list and explanation of where Shay locomotives are displayed or run.

9 Alishan Forestry and Railway Cultural Landscape [https://twh.boch.gov.tw/taiwan/intro.asp?id=7&lang=zh_tw]

阿里山林業暨鐵道文化景觀

PRIMARY Bureau of Cultural Heritage, Ministry of Culture (文化部文化資產局) accessed 2026-08-25

Official potential World Heritage-site page for the spiral, horseshoe, Z and Shay features, 6.25% gradient and 71.34 km landscape scope.

10 Alishan Forest Railway full-line opening on 6 July 2024 [<https://www.forest.gov.tw/news/0073945>]

阿里山林鐵113年7月6日全線通車

PRIMARY Forestry and Nature Conservation Agency, Ministry of Agriculture (農業部林業及自然保育署) accessed 2026-08-25

Official reopening announcement for typhoon damage, inspection, four daily services and the passenger opening date.

11 No. 42 tunnel completion ceremony [<https://afrch.forest.gov.tw/AllNews/0073708>]

林鐵42號隧道竣工典禮 祈願一路順遂

PRIMARY Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處)
accessed 2026-08-25

Official completion record for the collapsed No. 42 tunnel and the 1,106 m replacement tunnel.

12 Alishan Forest Railway [https://recreation.forest.gov.tw/Forestry/FR?typ=11&typ_id=0900041]

阿里山林業鐵路

PRIMARY Forestry and Nature Conservation Agency recreation site (林業及自然保育署台灣山林悠遊網)

accessed 2026-08-25

Government chronology for the 2019 important cultural landscape registration and the 2023/2024 reopening milestones.

13 Alishan Forest Railway service resumption on 5 October 2024 [<https://afrch.forest.gov.tw/AllNews/0074597>]

阿里山林鐵10月5日恢復營運

PRIMARY Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處)

accessed 2026-08-25

Official notice recording the October 2024 typhoon suspension, route inspection and 5 October service resumption.

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Station names, codes and sequence from TDX Transport Data eXchange, Ministry of Transportation and Communications, operators TRTC, NTMC, TYMC, and TMRT. Government open data. Retrieved 2026-08-24; source last updated 2016-11-23. The 12 Sanying station records absent from TDX are sourced directly to New Taipei Metro and government publications on their pages. Provenance.